

Generalization and Regularization

Making deep networks perform well on unseen data

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ES 667 · Deep Learning · IIT Gandhinagar

The generalization problem

The last lectures gave us a network that **trains**:

$$f_{\theta}(x) \rightarrow \hat{y}, \quad \mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(x_i), y_i)$$

$$g_t = \nabla_{\theta} \mathcal{L}(\theta_t), \quad \theta_{t+1} = \theta_t - \eta g_t$$

Backprop gives the gradient; the optimizer drives $\mathcal{L}_{\text{train}}$ down. We can now make the **training loss** small.

But is low training loss the goal?

Q. Does a low training loss guarantee a good model?

No.

We never cared about the training set. We care about **new, unseen** data.

Training loss is not the goal — test performance is.

Training loss — an empirical average over the data we happened to collect:

$$\mathcal{L}_{\text{train}(\theta)} = \frac{1}{n} \sum_{i=1}^n \ell(f_{\theta}(x_i), y_i)$$

Test loss — the expectation over the **true** data distribution p_{data} :

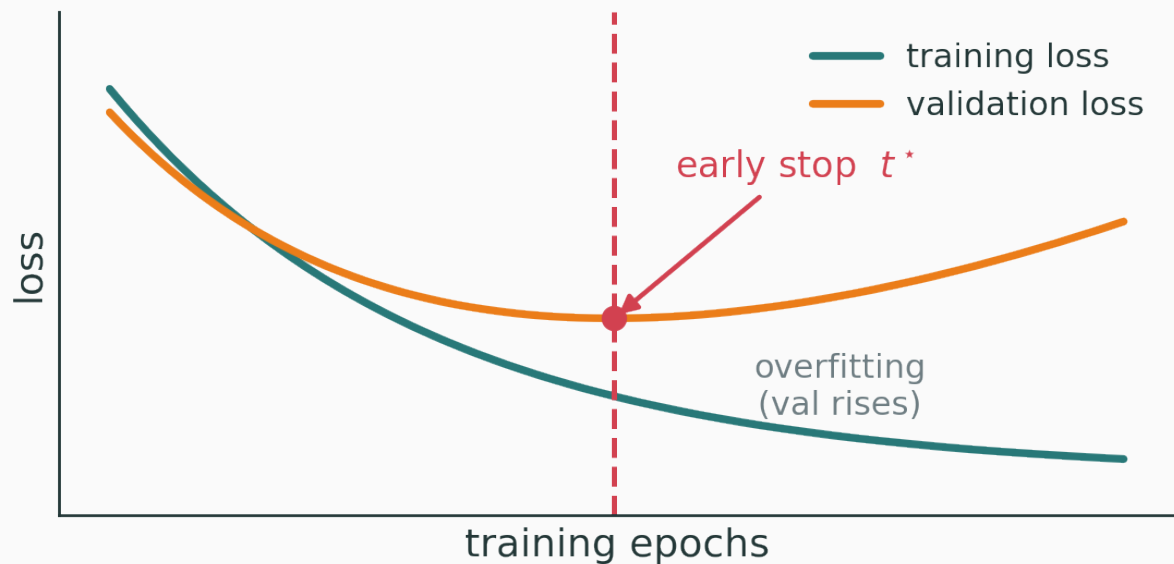
$$\mathcal{L}_{\text{test}(\theta)} = \mathbb{E}_{(x,y) \sim p_{\text{data}}} [\ell(f_{\theta}(x), y)]$$

We only ever **see finite data**; $\mathcal{L}_{\text{train}}$ is a noisy estimate of the quantity we actually want, $\mathcal{L}_{\text{test}}$.

The generalization gap

$$\underbrace{\mathcal{L}_{\text{test}} - \mathcal{L}_{\text{train}}}_{\text{generalization gap}}$$

Overfitting: training loss keeps falling while validation loss **turns up**.



Q. Give a big network **random** labels $y \rightarrow y_\pi$ (a fixed permutation). What happens?

- A high-capacity net still drives $\mathcal{L}_{\text{train}} \rightarrow 0$ — it **memorizes** the noise;
- but validation stays at chance: it learned **nothing generalizable**.

Fitting the training set is **not** evidence of learning. Capacity alone is not enough — we need **inductive bias** + an honest **validation** signal.

We have networks that train. Now we must stop them from **memorizing**.

Lever	What it controls
splits + early stopping	when to stop
weight decay / ℓ_2	size of the weights
dropout	co-adaptation of units
data augmentation / mixup	invariance, linearity
label smoothing	overconfidence
implicit regularization	SGD noise, architecture, init

Data splits and early stopping

Three splits, three jobs

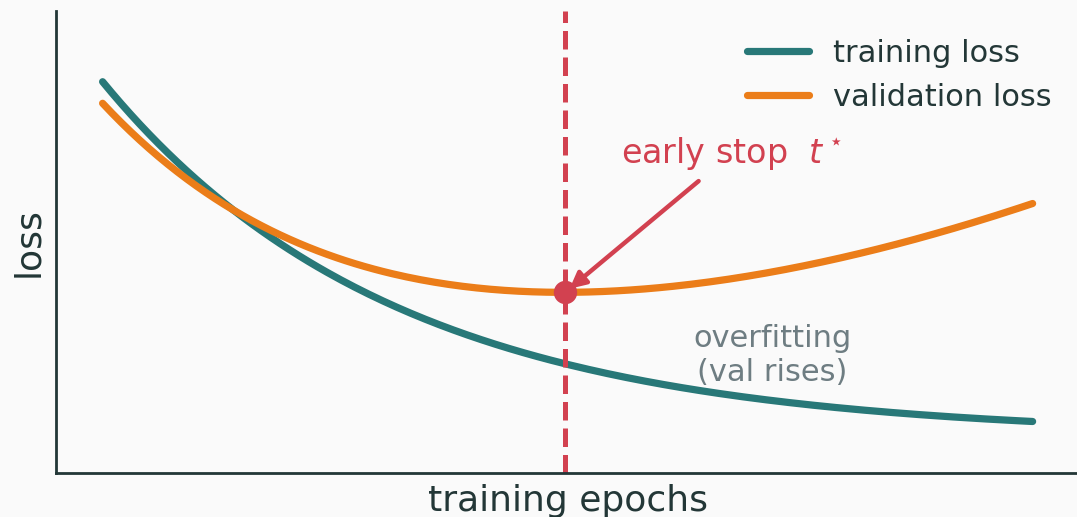
Split	Used to	Touches weights?
train	fit parameters θ	yes — via gradients
validation	select model / hyperparameters	no — only choices
test	final, unbiased estimate	no — used once

Never tune on the test set. Every peek leaks information and inflates your reported number.

Early stopping

Monitor validation loss each epoch; keep the **best** checkpoint, not the last:

$$t^* = \arg \min_t \mathcal{L}_{\text{val}}(\theta_t)$$



train keeps falling; validation is U-shaped — stop at the bottom.

Six epochs of training. Which checkpoint do we keep?

epoch	1	2	3	4	5	6
$\mathcal{L}_{\text{train}}$	1.20	0.80	0.45	0.32	0.28	0.25
$\mathcal{L}_{\text{val}}$	1.25	0.98	0.82	0.78	0.84	0.97

- Training loss is **lowest at epoch 6** — but that is not what we want;
- Validation loss bottoms out at **epoch 4** (0.78), then rises → overfitting.

keep θ_4 — the last epoch is the wrong choice

Early stopping as implicit regularization

Why does stopping early help?

- Weights **grow** over training; stopping early keeps them **small**;
- fewer effective updates $\Rightarrow$ a **simpler** function is explored;

Early stopping limits **effective capacity** without changing the architecture — a free, always-on regularizer. It is the cheapest one you have.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/double-descent

Sweep model capacity and training time and watch the train and validation curves separate — find the checkpoint where validation is lowest before the gap opens.

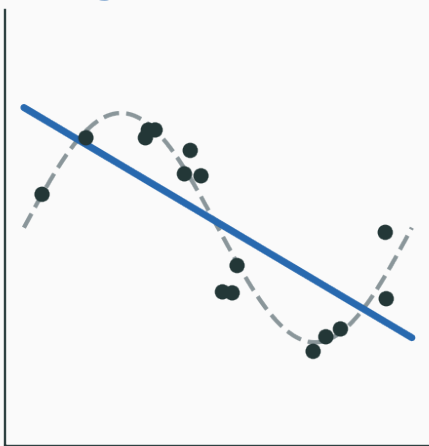
Q. You stop at the epoch with the lowest **training** loss. Why is your reported accuracy likely too optimistic?

Capacity, bias and variance

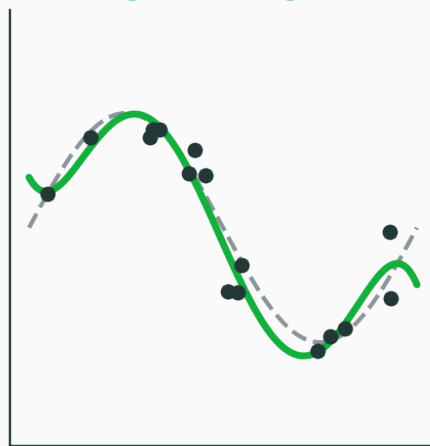
Three regimes

Regime	$\mathcal{L}_{\text{train}}$	$\mathcal{L}_{\text{val}}$	Diagnosis
underfit	high	high	too little capacity
good fit	low	low	capacity ~ right
overfit	low	high	memorizing train

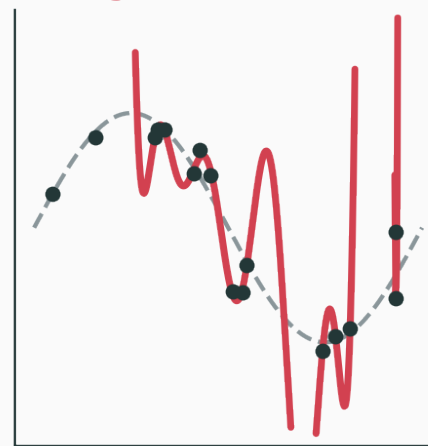
degree 1 — underfit



degree 5 — good



degree 15 — overfit



Expected test error splits into three pieces:

$$\mathbb{E}[(\text{error})^2] \approx \underbrace{\text{bias}^2}_{\text{too simple}} + \underbrace{\text{variance}}_{\text{too sensitive}} + \underbrace{\sigma^2}_{\text{irreducible noise}}$$

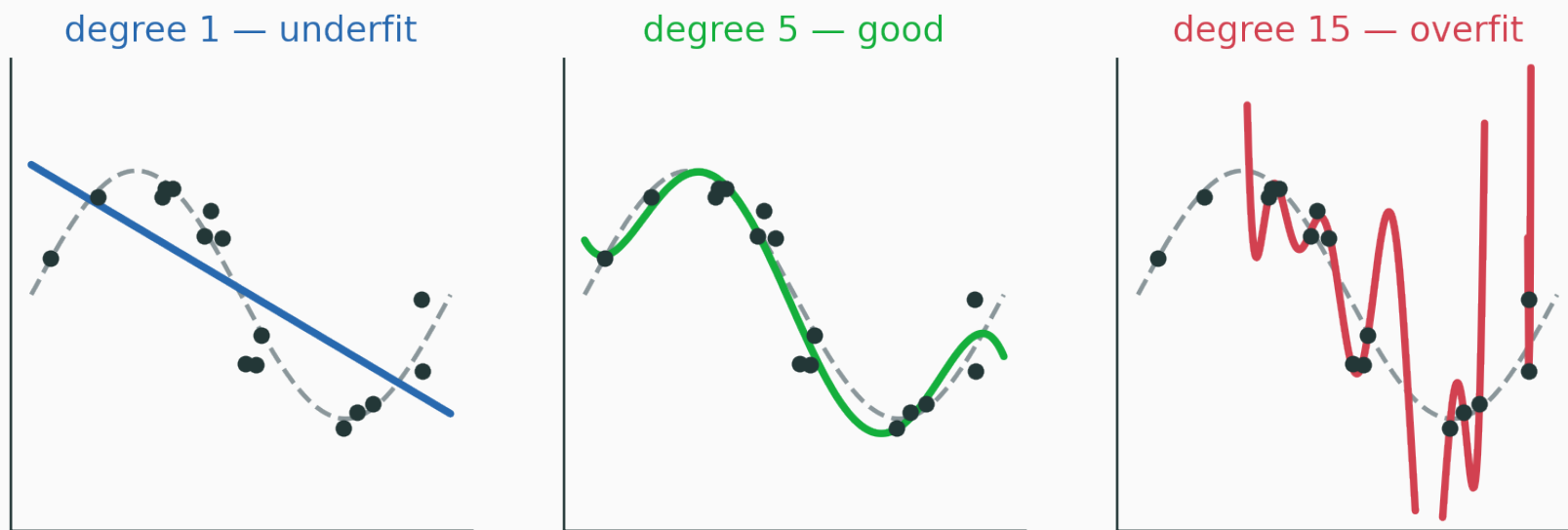
- **high bias** — the model is too simple; misses the true pattern (underfit);
- **high variance** — the model is too sensitive to the particular sample (overfit).

More capacity ↓ bias but ↑ variance. Classically we trade one against the other.

Worked example: polynomial fits

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Fit $y = \sin(2\pi x) + \varepsilon$ from noisy samples with polynomials of degree d :

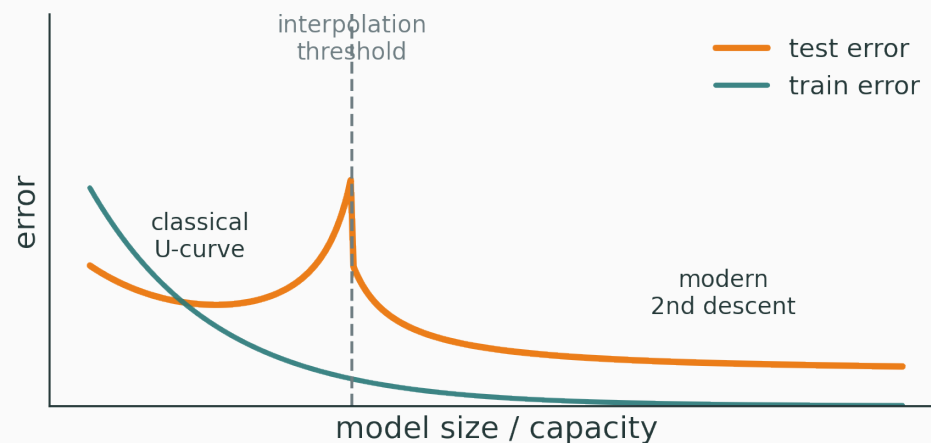


$d = 1$ underfits (high bias) · $d = 5$ fits · $d = 15$ overfits (high variance)

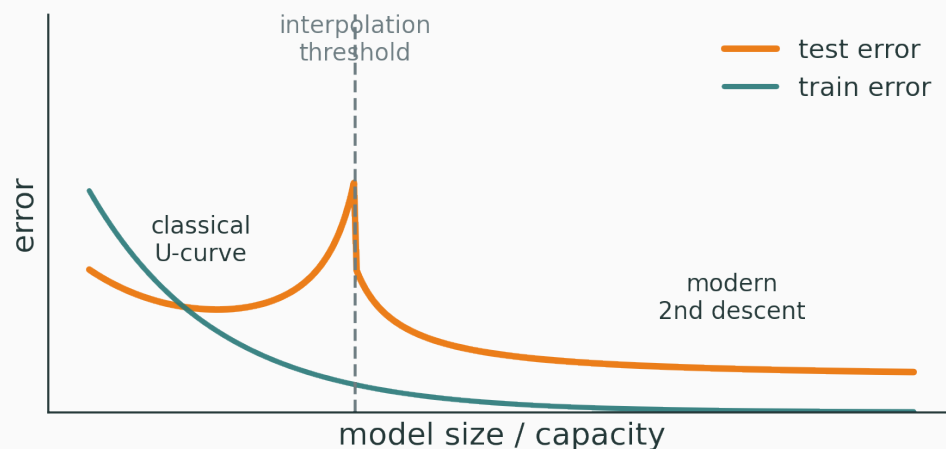
The deep-learning twist

Classical wisdom: **bigger model $\Rightarrow$ more overfitting**. Modern practice complicates this.

- huge over-parameterized nets often generalize **well**, not worse;
- what governs generalization: **optimization, data, architecture, implicit bias** — not just parameter count;



Double descent



Past the interpolation threshold, test error can **fall again** — the modern “second descent”.

Parameter count is a poor proxy for capacity in deep nets. (active research)

But **validation discipline is still non-negotiable** — it is how we know what is happening.

Weight decay and ℓ_2 regularization

Penalize large weights

Add a penalty on the **size** of the weights to the data loss:

$$\mathcal{L}_{\text{reg}}(\theta) = \mathcal{L}_{\text{data}}(\theta) + \lambda \|\theta\|^2$$

- $\lambda > 0$ — the **regularization strength** (a hyperparameter);
- large weights $\Rightarrow$ sharp, wiggly functions; the penalty prefers **smooth** ones.

ℓ_2 regularization $\Leftrightarrow$ a **Gaussian prior** on the weights: MAP estimation with $\theta \sim \mathcal{N}(0, \sigma^2 I)$ gives exactly this penalty.

Differentiate $\mathcal{L}_{\text{reg}} = \mathcal{L}_{\text{data}} + \lambda \|\theta\|^2$:

$$\nabla \mathcal{L}_{\text{reg}} = \nabla \mathcal{L}_{\text{data}} + 2\lambda \theta$$

Substitute into the SGD step:

$$\theta_{t+1} = \theta_t - \eta(\nabla \mathcal{L}_{\text{data}} + 2\lambda \theta_t)$$

$$\theta_{t+1} = (1 - 2\eta\lambda) \theta_t - \eta \nabla \mathcal{L}_{\text{data}}$$

a multiplicative shrinkage $(1 - 2\eta\lambda)$ before the gradient step

Why “weight decay”?

Each step first **shrinks** the weight toward zero, then applies the gradient:

$$\theta_t \mapsto (1 - \eta\lambda') \theta_t$$

Absent any gradient, weights **decay** geometrically toward 0 — hence the name. ℓ_2 regularization and weight decay are the same mechanism (for SGD).

$\theta = 5$, gradient $g = 3$, learning rate $\eta = 0.1$, decay $\lambda' = 0.2$: **Without** weight decay:

$$\theta^+ = \theta - \eta g = 5 - 0.1 \cdot 3 = 4.7$$

With weight decay $\theta^+ = (1 - \eta\lambda')\theta - \eta g$:

$$\theta^+ = (1 - 0.1 \cdot 0.2) \cdot 5 - 0.1 \cdot 3 = 0.98 \cdot 5 - 0.3 = 4.6$$

the extra -0.1 is the shrinkage pulling the weight toward zero.

For **SGD**, adding $\lambda\|\theta\|^2$ to the loss $\equiv$ weight decay (up to scaling). For **Adam** they **differ**: the $2\lambda\theta$ term gets divided by $\sqrt{\hat{v}}$ like any gradient — so decay is **distorted** per coordinate. Loshchilov & Hutter **decoupled** it — apply decay directly to the weights:

$$\theta \mapsto \text{AdamStep}(\theta), \quad \theta \mapsto (1 - \eta\lambda) \theta$$

use AdamW, not Adam with naive ℓ_2

When (and when not) to decay

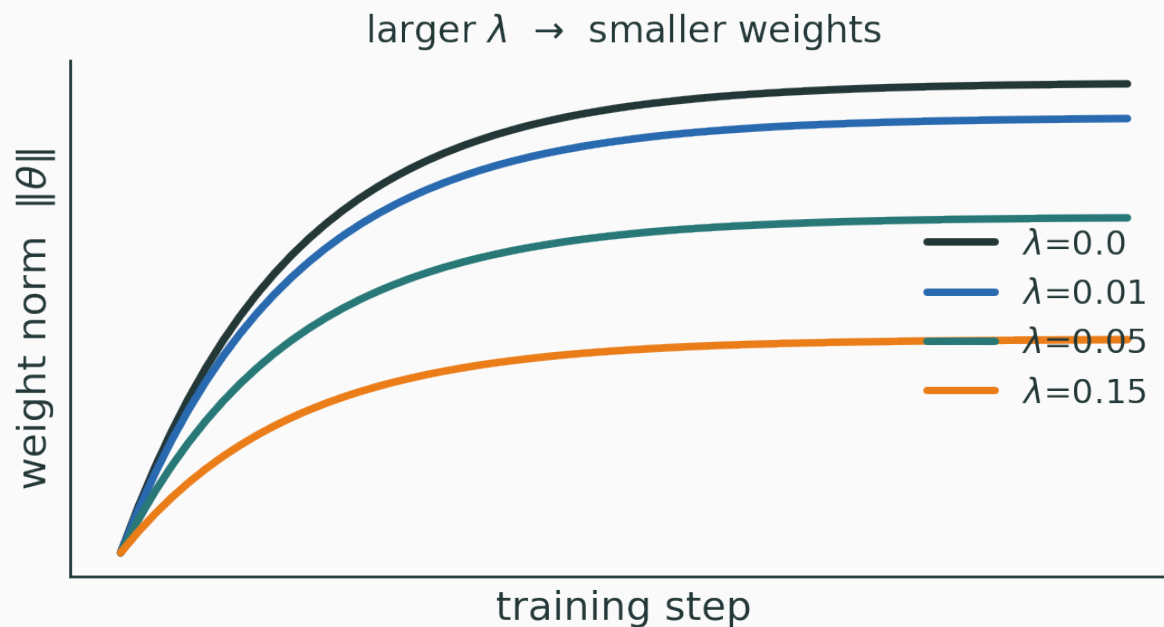
- Use it almost everywhere: **MLPs, CNNs, Transformers**, large models;
- typical λ : 10^{-4} to 10^{-2} (tune on validation);
- **do not** decay **biases**, **norm** scale/shift, and often not **embeddings** — decaying them helps nothing and can hurt.

Rule of thumb: decay the **weight matrices**, leave the **one-dimensional** parameters alone.

Weight decay shrinks the weights

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Weight norm $\|\theta\|$ over training for several λ :



larger $\lambda \Rightarrow$ smaller final weights $\Rightarrow$ smoother function.

INTERACTIVE

(to build) — sweep λ and watch the weight norm shrink and the decision boundary smooth out; too large and the model underfits.

Q. Doubling η and halving λ leaves $\eta\lambda$ fixed. Does weight decay behave the same? (Careful — the **data** step changed too.)

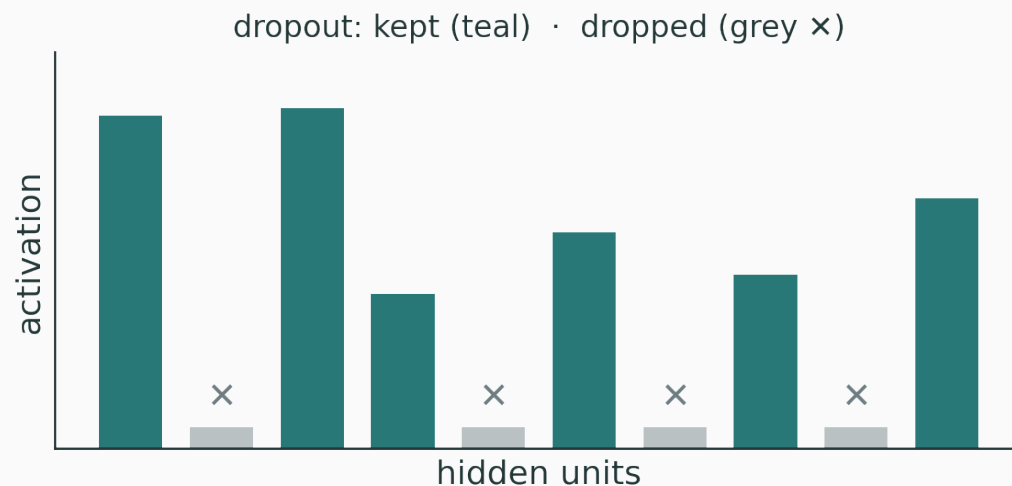
Dropout

The idea

During training, randomly **drop** each hidden unit with probability $1 - p$:

$$m_j \sim \text{Bernoulli}(p), \quad \tilde{h} = m \odot h$$

- a different random **subnetwork** is trained on every minibatch;
- units cannot rely on any single partner $\Rightarrow$ less **co-adaptation**.



Dropping units shrinks the expected activation. **Rescale** by $\frac{1}{p}$ during training:

$$\tilde{h} = \frac{m \odot h}{p}$$

Then the expected activation is **unchanged**:

$$\mathbb{E}[\tilde{h}_j] = \frac{1}{p} \cdot (p \cdot h_j + (1 - p) \cdot 0) = h_j$$

no test-time rescaling needed — at test time just use the full network

$h = [2, 4, 6, 8]$, keep prob $p = 0.5$, sampled mask $m = [1, 0, 1, 0]$:

$$\tilde{h} = \frac{m \odot h}{p} = \frac{[2, 0, 6, 0]}{0.5} = [4, 0, 12, 0]$$

Check the expectation (each unit kept w.p. 0.5, scaled by 2):

$$\mathbb{E}[\tilde{h}] = 0.5 \cdot \left(\frac{h}{0.5} \right) = h = [2, 4, 6, 8]$$

surviving units are **doubled**, so the layer's expected output is preserved.

Why it helps: model averaging

Each mask defines a **different subnetwork**; over training we fit exponentially many. At test time the full network **approximates an average** of all these subnetworks.

An **ensemble-like** effect from one model — but this is an intuition, not an exact Bayesian average. Do not overclaim it.

- Works best on **fully-connected** layers and **classifier heads**;
- less useful alongside strong **augmentation** and **normalization** (which already regularize);
- adds gradient noise $\Rightarrow$ can **slow optimization**, needs more epochs;
- tune the **keep** rate $p \approx 0.5\text{--}0.8$ (i.e. drop 20–50%).

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↗ nipunbatra.github.io/interactive-articles/dropout-playground

Toggle dropout on a small MLP, sweep the rate, and watch training vs validation loss — see how too much drop underfits and too little overfits.

Data augmentation

Generate new **label-preserving** examples from the ones you have:

$$(x, y) \mapsto (T(x), y), \quad T \text{ preserves the label}$$

Domain	Transforms T
images	crop, flip, rotate, color jitter, noise
audio / time series	time-shift, mask spans, add noise
text	paraphrase, synonym swap, back-translation

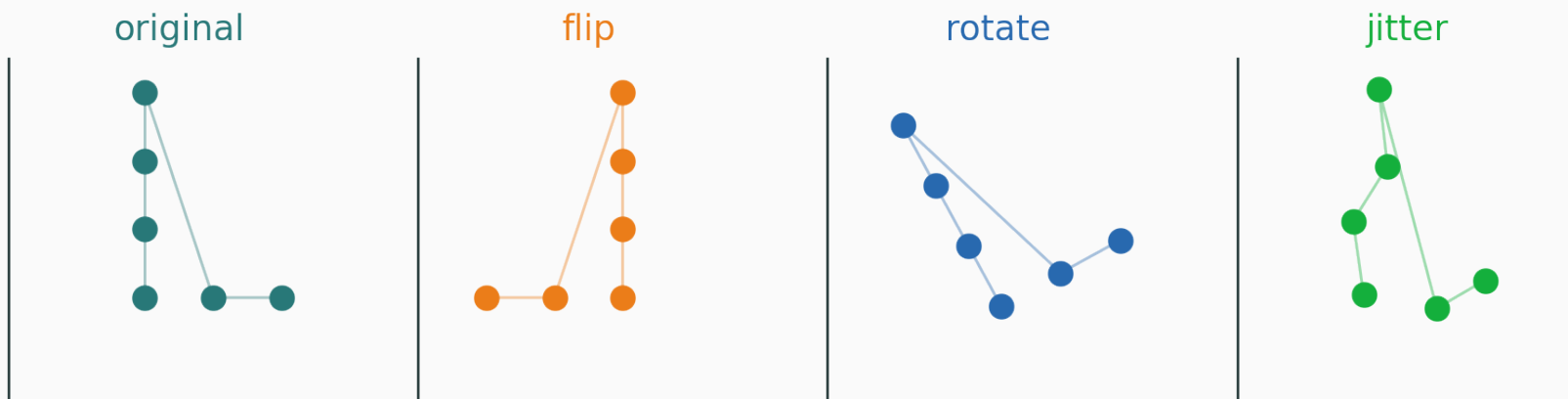
Augmentation teaches invariance

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If T preserves the label, we want the model to be **invariant** to it:

$$f_{\theta}(x) \approx f_{\theta}(T(x))$$

same label under label-preserving transforms



same object, many views — the network learns to ignore the nuisance transform.

Bad augmentation encodes wrong knowledge

Augmentation **injects domain knowledge** — the wrong T injects the wrong knowledge:

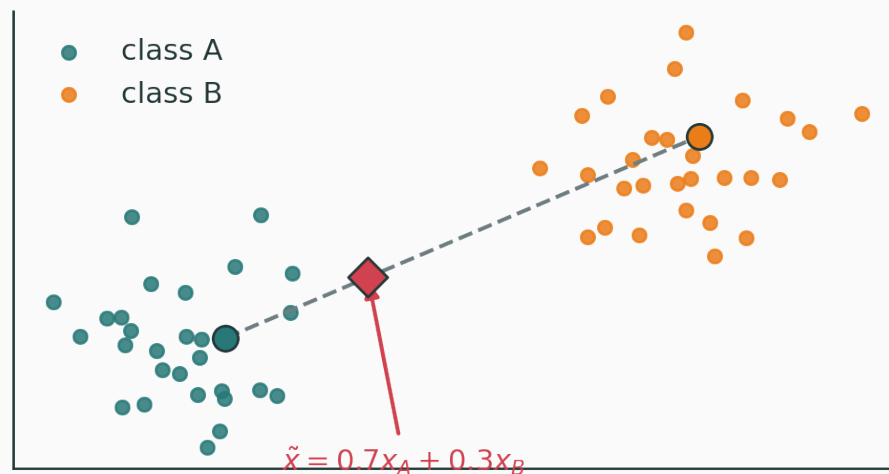
- flipping a **6** into a **9** — the label changed;
- **time-reversing** speech or a heartbeat — no longer valid;
- **color** jitter when **color is the label** (ripe vs unripe fruit);
- **cropping out** the object you are trying to classify.

T must preserve the label. Choosing T **is** a modelling decision about what should not matter.

Mixup: interpolate examples

Blend two examples **and** their labels with $\lambda \sim \text{Beta}(\alpha, \alpha)$:

$$\tilde{x} = \lambda x_i + (1 - \lambda)x_j, \quad \tilde{y} = \lambda y_i + (1 - \lambda)y_j$$



convex combinations encourage **linear** behaviour between examples.

$x_A = \text{cat}, y_A = [1, 0]; x_B = \text{dog}, y_B = [0, 1]; \text{mix } \lambda = 0.7:$

$$\tilde{x} = 0.7 x_A + 0.3 x_B$$

$$\tilde{y} = 0.7[1, 0] + 0.3[0, 1] = [0.7, 0.3]$$

Train with the **soft** cross-entropy against this target:

$$\ell = - \sum_k \tilde{y}_k \log p_k = -0.7 \log p_1 - 0.3 \log p_2$$

the network is asked to be 70% cat, 30% dog — no hard boundary jump.

INTERACTIVE

(to build) — apply transforms and mixup to a 2-D dataset and watch the decision boundary become smoother and more invariant.

Q. Why does mixup improve **calibration** as well as accuracy?

Label smoothing

Hard labels push to extremes

With a one-hot target, cross-entropy keeps pushing $p_y \rightarrow 1$:

$$\ell = -\log p_y \implies \text{logits} \rightarrow \pm\infty$$

The model is rewarded for **ever more extreme** confidence — even on ambiguous or mislabelled examples. This encourages **overconfidence**.

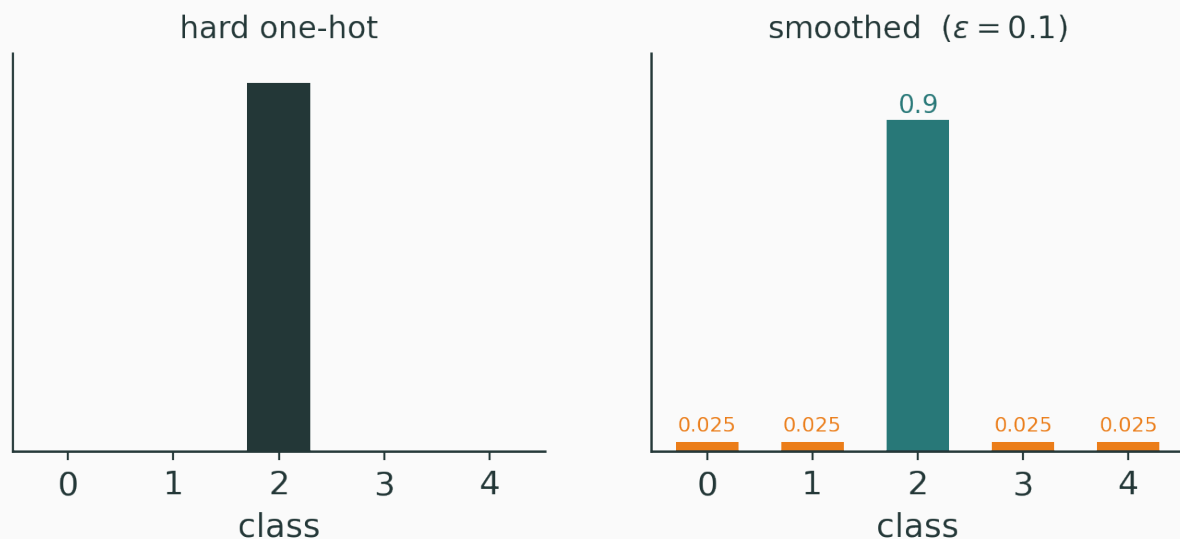
Softening the target

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Move a little mass ε off the correct class onto the rest:

$$y_k^{\text{smooth}} = \begin{cases} 1 - \varepsilon & \text{if } k = y \\ \varepsilon / (K - 1) & \text{otherwise} \end{cases}$$

Equivalently, mix the one-hot with a uniform distribution over the **other** classes.



$K = 5$ classes, smoothing $\varepsilon = 0.1$, correct class = 2:

$$\text{correct : } 1 - \varepsilon = 0.9, \quad \text{each other : } \frac{\varepsilon}{K - 1} = \frac{0.1}{4} = 0.025$$

$$y^{\text{smooth}} = [0.025, 0.025, 0.9, 0.025, 0.025]$$

still sums to 1; the target is achievable with **finite** logits.

Why it helps (and when it hurts)

- **discourages extreme confidence** — logits stay bounded;
- improves **calibration** (predicted probabilities match accuracy);
- adds **robustness to noisy labels** and often improves generalization;

Can **hurt** when you need exact confidences — e.g. **knowledge distillation** or when downstream code reads calibrated probabilities.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/calibration

Sweep ε and watch the reliability diagram — see over-confident predictions pull back toward the diagonal as smoothing increases.

Noise and implicit regularization

Noise is a regularizer

Many effective regularizers work by **injecting noise** during training:

- **SGD** minibatch sampling — noisy gradients;
- **dropout** — noisy activations;
- **augmentation** — noisy inputs;
- **stochastic depth, label / input noise**.

Noise prevents the network from building **brittle**, precise dependencies — it must be robust to the perturbation.

Explicit vs implicit regularization

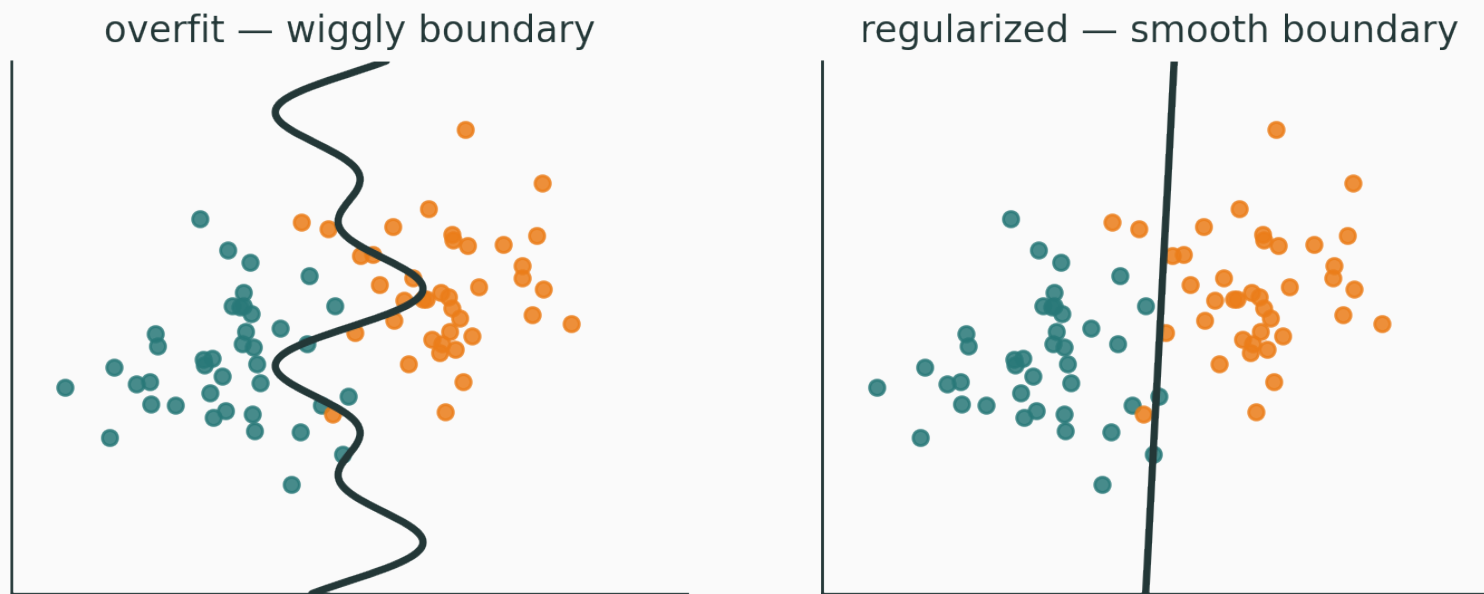
Explicit (a term / operation)	Implicit (a side effect)
$+\lambda\ \theta\ ^2$ penalty	SGD gradient noise
dropout	architecture / optimizer bias
data augmentation	early stopping
label smoothing	normalization, initialization

Regularization is **not only a penalty term**.

Regularization changes the learned function

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Two models can both fit the training set to $\mathcal{L}_{\text{train}} \approx 0$ — yet differ:



regularization **picks** the smoother boundary — smaller norm, more invariance, less confidence, better val.

A practical workflow

Q. Both training and validation loss are **high**. Add dropout?

No — that is underfitting.

Fix the fit first: **bigger model**, **train longer**, **better LR**, **less augmentation**, **check the labels**.

Adding regularization to an underfit model makes it **worse**. Regularize only once you can **overfit**.

Then: regularize an overfit model

Training loss **low**, validation loss **high** $\Rightarrow$ overfitting. Now reach for:

Lever	First choice when...
more data / augmentation	cheap to collect more or transform
weight decay	almost always — near-free
dropout	MLP / classifier head
early stopping	always on
smaller model	data is truly limited
label smoothing	over-confident / noisy labels

A regularization dashboard

Track more than the loss — watch **what** is being controlled:

- $\mathcal{L}_{\text{train}}$ **and** $\mathcal{L}_{\text{val}}$ (and the gap between them);
- $\text{acc}_{\text{train}}$ **and** acc_{val} ;
- weight norm $\|\theta\|$;
- confidence $\max_k p_k$ and **calibration error**.

If the **gap** widens while train keeps improving, you are overfitting — turn a regularization knob.

Model	$\text{acc}_{\text{train}}$	acc_{val}	Action
A	65%	63%	underfit $\rightarrow$ add capacity , train longer
B	99%	72%	overfit $\rightarrow$ regularize / augment / early-stop

- **A**: small gap, both low $\Rightarrow$ the model **cannot fit** — do **not** regularize;
- **B**: large gap $\Rightarrow$ the model **memorized** — rein it in.

Practical defaults

Setting	Default recipe
images (CNN)	augmentation + weight decay + early stopping
MLP / tabular	weight decay + dropout + early stopping
Transformer	AdamW + dropout + (label smoothing) + aug
small data	transfer learning + strong validation

Start simple, add one knob at a time, and let **validation** decide whether it helped.

Summary

One table: every method

Method	What it controls
early stopping	effective complexity — when to stop
weight decay	parameter norm $\ \theta\ $
dropout	co-adaptation / activation noise
data augmentation	invariance to nuisance transforms
mixup	linearity between examples
label smoothing	overconfidence / calibration
validation split	model selection — the judge

Optimization **reduces training loss**.
Regularization **improves unseen-data performance**.

validation tells us whether it is actually helping.

Optimization reduces train loss; regularization improves generalization; validation is the judge.

Next: **CNNs** — why fully-connected nets are inefficient for images.